Research Revision Supporting Reports

Student Academic Risk Classification
1 November 2026

The original experiment reproduces. The revised evidence remains retrospective and exploratory, with unresolved author declarations and independent validation requirements. This document contains the technical audit, experiment record, reproduction instructions and submission checklist. Detailed numerical outputs remain in the accompanying CSV files.

Technical Audit

Review date: 1 November 2026. All nine original PDF pages and the available source, saved models, predictions, metadata, tests, and current interface were inspected. Severity denotes impact on claim validity; it is not a publication acceptance prediction.

A01 Early detection title

Severity: Critical. Title implies a verified prediction cutoff; no predictor timestamps exist.

Action and evidence: Renamed retrospective classification; revised abstract, objectives and conclusion.

Residual requirement: Needs new timestamped cohort for prospective claim.

A02 Absences availability

Severity: High. Aggregate absences may span the outcome period.

Action and evidence: Retuned no-absences models under the same nested folds.

Residual requirement: Other feature timestamps remain unknown.

A03 Small holdout

Severity: High. Only 26 positive outcomes produce wide rate uncertainty.

Action and evidence: Wilson recall intervals and conditional paired bootstrap intervals added.

Residual requirement: Larger independent cohorts required.

A04 Tuning optimism

Severity: High. Five-fold maximum scores reuse the folds that choose settings.

Action and evidence: Added 5-fold x 2-repeat outer validation with complete inner search on original training rows.

Residual requirement: Dependent folds are not independent cohorts.

A05 Known holdout

Severity: High. Revision was designed after original test results were known.

Action and evidence: Extensions labeled exploratory; fixed original row IDs, no threshold/default replacement.

Residual requirement: Fresh validation remains necessary.

A06 Objective baseline

Severity: High. Always-At-Risk exceeds both original models on test F2.

Action and evidence: Retained baseline; explicitly interpret precision, recall and workload.

Residual requirement: No institution-specific cost or capacity elicitation.

A07 Historical rule baseline

Severity: Medium. Original study lacked a past-failure rule.

Action and evidence: Added failures > 0 rule without tuned cutoff.

Residual requirement: Timing of recorded history still requires confirmation.

A08 Probability benchmark

Severity: Medium. Hard constant baselines are weak probability comparators.

Action and evidence: Added training prevalence constant for Brier comparison.

Residual requirement: Hard heuristic scores are not probabilities.

A09 Model diversity

Severity: Medium. Two models alone do not test nonlinear ensemble behavior.

Action and evidence: Added bounded RF and GB searches using same partitions and scoring.

Residual requirement: Search budgets differ; no universal family ranking.

A10 No calibration evidence

Severity: High. Brier and displayed percentages do not establish calibration.

Action and evidence: Added nested-search sigmoid calibration ensembles and fixed-bin reliability diagnostics.

Residual requirement: Small test bins; ensemble effect confounds pure recalibration effect.

A11 Subgroup fairness

Severity: High. No sex, address or school error assessment existed.

Action and evidence: Added denominators, recall and FPR with Wilson intervals.

Residual requirement: No fairness certification, causal analysis or mitigation.

A12 Sensitive predictors

Severity: Medium. Private, health and proxy inputs create use concerns.

Action and evidence: Retuned reduced 15-feature sensitivity scenario.

Residual requirement: Removal does not eliminate proxies or establish fairness.

A13 Prior-failure dependence

Severity: Medium. Unclear whether other information adds predictive value.

Action and evidence: Added no-failures ablation and simple-rule comparison.

Residual requirement: No causal effect inferred.

A14 Threshold selection

Severity: Medium. F2 search at 0.5 may hide workload trade-offs.

Action and evidence: Added descriptive 0.1-0.9 test sensitivity curves.

Residual requirement: No test-based threshold tuning; capacity study pending.

A15 Paired uncertainty

Severity: Medium. Ranking differences had point estimates only.

Action and evidence: 2,000 paired stratified bootstraps; exact LR-DT McNemar test.

Residual requirement: Intervals condition on fixed models/prevalence, omit cluster/retraining uncertainty.

A16 Fold independence

Severity: Medium. Repeated folds overlap in fit rows and students.

Action and evidence: Means and sample SD only; no naive fold-based significance test.

Residual requirement: External replication remains needed.

A17 Importance bias

Severity: Medium. Training impurity can favor split-rich inputs.

Action and evidence: Added raw-feature AP permutation on outer validation folds.

Residual requirement: Correlation and out-of-distribution permutations remain limitations.

A18 Explanation stability

Severity: Medium. One fitted coefficient/tree ranking is unstable on small data.

Action and evidence: Saved per-fold LR coefficients and validation permutation values.

Residual requirement: No causal or intervention interpretation.

A19 Odds ratio units

Severity: Medium. Numeric and categorical coefficients have different units.

Action and evidence: Clarified per-training-SD effects and omitted categorical references.

Residual requirement: No conventional unpenalized inferential CI claimed.

A20 Ordinal codes

Severity: Medium. LR treats ordinal steps as equally spaced.

Action and evidence: Preserved actual encoding and stated assumption.

Residual requirement: Alternative categorical/spline coding is a possible future study.

A21 External validation

Severity: High. No independent school/cohort evaluation.

Action and evidence: Scope limited; public future benchmark suitability documented.

Residual requirement: Requires compatible independent data; not repairable with random resampling.

A22 Temporal validation

Severity: High. Random split does not test future cohorts.

Action and evidence: Explained cutoff and cohort validation requirements; assessed OULAD compatibility.

Residual requirement: New temporal data needed; OULAD is a separate task.

A23 School clustering

Severity: Medium. Shared school/class context may overstate independence.

Action and evidence: Reported assumptions; school subgroup counts retained.

Residual requirement: Only two schools and no class IDs; no cluster-robust inference claimed.

A24 Zero grades

Severity: Medium. 38 zero grades may mix absence/withdrawal/achievement.

Action and evidence: Retained original target and explicitly reported ambiguity.

Residual requirement: Administrative meaning needs source confirmation.

A25 Dataset identity

Severity: Medium. UCI summary has 649 language records versus 395 mathematics rows.

Action and evidence: Verified actual shape, hash, labels, missingness and full-row duplicates.

Residual requirement: No mathematics/language pooling or false external validation.

A26 Novelty

Severity: Medium. Two standard algorithms and an app offer limited algorithmic contribution.

Action and evidence: Reframed as reproducibility and sensitivity audit with bounded claims.

Residual requirement: Venue fit and sufficient significance require editorial assessment.

A27 Literature currency

Severity: Medium. Original bibliography mostly predates 2023.

Action and evidence: Added six relevant 2023-2026 works; DOI and official-source records saved.

Residual requirement: Focused update, not a systematic literature review.

A28 Environment claim

Severity: Medium. Paper says Python 3.12.14 while current environment is 3.12.10.

Action and evidence: Reported observed interpreter and retained historical metadata separately.

Residual requirement: Historical run environment not retroactively altered.

A29 Prototype privacy

Severity: Medium. Current interface requests a name unnecessary for inference.

Action and evidence: Inspected session/code behavior; recommended data minimization.

Residual requirement: App is not modified or represented as validated deployment.

A30 Unsupported declarations

Severity: High. Funding/conflicts/ethics/repository placeholders were unresolved.

Action and evidence: Removed placeholders and invented acknowledgment; tracked author actions separately.

Residual requirement: Authors must provide actual declarations and archive URL.

A31 Authorship and biographies

Severity: High. Author identity must remain and biographies are required by Access.

Action and evidence: Names/order/affiliation/correspondence preserved; no biographies invented.

Residual requirement: All authors must supply approved biographies and verify identity details.

A32 AI disclosure

Severity: Medium. Original AI declaration was a placeholder.

Action and evidence: Factual acknowledgment describes AI assistance and cites the system.

Residual requirement: Authors must approve disclosure and verify all assisted material.

A33 Formatting and equations

Severity: Medium. Original PDF had large blank areas and was not the Access template.

Action and evidence: Built editable DOCX from official template; added native equations, numbered figures/tables.

Residual requirement: Final Word/PDF layout and content checked; submission still awaits author actions.

A34 Unknown categories

Severity: Low. Ignored one-hot levels can encode like a reference category.

Action and evidence: Documented pipeline behavior and inspected application checks.

Residual requirement: Unknown cohort categories and range drift require deployment validation.

What was already correct

The original manuscript already described retrospective scope, no test threshold tuning, absence of fairness/calibration analysis, and the Always-At-Risk F2 limitation. These were not newly discovered code defects. The title remained inconsistent with its careful text. Pipeline isolation, grade exclusion, class-1 orientation, test confusion matrices, AP definition, class weighting, and coefficient units were implemented correctly. The original 39-column encoding and saved probabilities reproduced.

Experiment Report

All numbers below are executed results. Extensions are exploratory. The original test cohort was already known; no new holdout is claimed.

Protocol

Outer evaluation: 5-fold stratified, 2 repeats, seed 2026, on original 316 training records only. Inner search: 5-fold stratified shuffled seed 42, F2 then recall then ROC-AUC tie-break.

logistic regression: C=[0.01, 0.1, 1.0, 10.0]; penalty=['l1', 'l2']; class_weight=[None, 'balanced']

decision tree: criterion=['gini', 'entropy']; max_depth=[2, 3, 4, 5, 6]; min_samples_leaf=[5, 10, 20]; class_weight=[None, 'balanced']

random forest: max_depth=[3, None]; min_samples_leaf=[10]; class_weight=[None, 'balanced']

gradient boosting: learning_rate=[0.03, 0.1]; max_depth=[1, 2]

Calibration: sigmoid, ensemble=True, 3 stratified calibration folds seed 73; nested 5-fold hyperparameter searches within each fitting fold. Bootstrap: 2000 class-stratified paired resamples seed 2701; fixed fitted models; percentile 95% intervals.

Preserved original experiment

Both original models reproduce saved probabilities to maximum absolute difference 1.1102230246251565e-16. The original DT default and 0.5 threshold are retained. See results/reproduction_checks.json and frozen_split.json.

Executed analyses

Original 16-candidate LR and 60-candidate DT searches; four-candidate RF and GB searches; ten nested outer folds with four model families and four baselines; LR/DT retuned ablations for no absences, reduced personal attributes, and no failures; nested-search sigmoid calibration ensembles; 2,000 paired class-stratified bootstraps; exact McNemar; three subgroup slices; threshold curves; raw-feature outer-validation permutation and coefficient stability.

Actual held-out results

Model	Accuracy	Precision	Recall	F1	F2	AUC	AP	Brier	Flags
logistic_regression	0.671	0.500	0.692	0.581	0.643	0.744	0.598	0.207	36
decision_tree	0.620	0.450	0.692	0.545	0.625	0.693	0.569	0.216	40
random_forest	0.722	0.583	0.538	0.560	0.547	0.723	0.676	0.209	24
gradient_boosting	0.734	0.647	0.423	0.512	0.455	0.721	0.564	0.197	17
logistic_regression__no_absences	0.658	0.486	0.654	0.557	0.612	0.735	0.628	0.209	35
decision_tree__no_absences	0.608	0.444	0.769	0.563	0.671	0.701	0.530	0.236	45
logistic_regression__reduced_sensitive	0.671	0.500	0.538	0.519	0.530	0.698	0.571	0.218	28
decision_tree__reduced_sensitive	0.684	0.524	0.423	0.468	0.440	0.599	0.448	0.250	21
logistic_regression__no_failures	0.595	0.432	0.731	0.543	0.642	0.638	0.478	0.242	44
decision_tree__no_failures	0.557	0.415	0.846	0.557	0.701	0.689	0.471	0.221	53
logistic_regression sigmoid	0.696	1.000	0.077	0.143	0.094	0.748	0.607	0.197	2
decision_tree sigmoid	0.696	1.000	0.077	0.143	0.094	0.678	0.599	0.202	2
majority	0.671	0.000	0.000	0.000	0.000	0.500	0.329	0.329	0
always_at_risk	0.329	0.329	1.000	0.495	0.710	0.500	0.329	0.671	79
prevalence	0.671	0.000	0.000	0.000	0.000	0.500	0.329	0.221	0
prior_failure	0.734	0.647	0.423	0.512	0.455	0.655	0.464	0.266	17

Nested outer-fold results

The ten fold scores overlap and are dependent. Mean and sample SD are descriptive; they are not test-set results or independent-replicate confidence intervals.

Feature subset	Model	F2 mean	F2 SD	AUC mean
all	always_at_risk	0.7103	0.0061	0.5000
all	decision_tree	0.5385	0.1127	0.6186
all	gradient_boosting	0.3383	0.0937	0.6462
all	logistic_regression	0.4985	0.1403	0.6657
all	majority	0.0000	0.0000	0.5000
all	prevalence	0.0000	0.0000	0.5000
all	prior_failure	0.4220	0.1365	0.6386
all	random_forest	0.4681	0.1455	0.6650
no_absences	decision_tree	0.4955	0.0944	0.6217
no_absences	logistic_regression	0.5033	0.1185	0.6587
no_failures	decision_tree	0.5064	0.0915	0.5598
no_failures	logistic_regression	0.5252	0.1033	0.6103
reduced_sensitive	decision_tree	0.5680	0.1228	0.6419
reduced_sensitive	logistic_regression	0.5296	0.1112	0.6776

Statistical comparisons

LR correct DT wrong = 15; DT correct LR wrong = 11; exact two sided p = 0.557197093963623; exploratory = True.

Paired differences and percentile limits are in paired_differences.csv. Metric intervals condition on observed class counts and fixed fitted models. No cluster adjustment, model-refitting bootstrap or multiplicity-adjusted confirmatory significance claim is made.

Interpretation

F2 rewards broad referral; compare both precision and workload with Always-At-Risk. Rank metrics and Brier measure different properties. The prevalence baseline is the appropriate constant probability comparison. Calibration ensembles change base fits and averaging; their performance change is not solely a calibration effect. Ablations are predictive sensitivity tests. Small subgroup denominators prevent certification of fairness.

Not performed

No external or prospective cohort validation, prediction-cutoff reconstruction, OULAD training, Portuguese-subject transfer, operational intervention, teacher usability trial, fairness mitigation, Explainable Boosting Machine, SHAP study, or ordinal recoding experiment was performed. These are neither implicit successes nor fabricated pending results.

Output map

heldout_metrics.csv includes all completed variants; nested_fold_metrics.csv and nested_predictions.csv contain the repeated nested runs; subgroups.csv includes denominators and Wilson intervals; calibration_bins.csv

includes bin counts and observed-rate intervals; bootstrap_intervals.csv contains eight metric intervals for the four model families. Every figure is generated by make_figures.py from those outputs.

Reproducibility Report

Run date: 1 November 2026, Asia/Calcutta. Dataset: UCI mathematics file, CC BY 4.0. No synthetic data. Original source and artifacts remain in their original directories; new results are written only under publication.

Observed environment

Python 3.12.10. Windows-11-10.0.26200-SP0.

numpy 2.2.6; pandas 2.3.3; scipy 1.16.2; scikit-learn 1.7.2; matplotlib 3.10.7; joblib 1.5.2.

Recorded experiment elapsed time: 992.0 seconds.

Commands from the project root

```
.venv/Scripts/python.exe publication/source/run_experiments.py
.venv/Scripts/python.exe publication/source/make_figures.py
.venv/Scripts/python.exe -m pytest -q --basetemp=publication/review/pytest-new-run
```

Use a fresh, nonexistent test temporary directory inside the workspace and set TEMP and TMP to a writable workspace location when the restricted default Windows temp directory is unavailable. The first test attempt had three temp-directory setup errors; the second used a writable directory and passed all 27 tests, with the expected historical-source warning.

Artifact authoring

The bundled Codex Python is used for DOCX authoring and PDF rasterization. It requires python-docx, pypdfium2 and Pillow. Run source/build_documents.py to rebuild both DOCXs and Markdown; run source/export_word.ps1 with Microsoft Word installed to export the PDFs; run source/render_pdf.py for the page images used in QA. The packaged render_docx.py was attempted and failed because LibreOffice is unavailable; Word was used as the native fallback.

Portable reproduction

The source bundle includes a standalone project copy under reproducibility/project, with src/, data/, original model metadata and predictions needed for exact comparison. The runner detects that it is in that copied project using its source location. Place the publication source directory in project/publication/source, as packaged, and run from project. Install the scientific requirements in reproducibility/requirements-experiments.txt in an isolated Python 3.12 environment. The large original app lockfile is retained separately for app reconstruction; its historic package versions may require an appropriate index.

Protocol details

Read results/protocol.json for grids, seeds, nested split logic, calibration and bootstrap design. frozen_split.json preserves the original 316/79 row IDs. nested_folds.json preserves every outer fit/validation ID; inner folds are deterministic from their fitting row order and seed 42. Pipeline preprocessing and weighting are refitted within fitting partitions. No test-based threshold, feature or final-model choice is made.

Integrity checks

SHA256SUMS.json records final deliverables, source files, data and experiment artifacts. Reproduction checks assert row order and original probabilities before extensions run. No external data or network connection is necessary for experiments; verify_references.py is the optional network-dependent bibliography audit. Joblib artifacts must be loaded only from this trusted generated package.

Hardware and repeatability

All estimator fits use n_jobs=1. Runtime is recorded in environment.json, not treated as a portable performance benchmark. Floating-point differences across libraries or platforms may occur. The scripted probability comparison uses absolute and relative tolerance 1e-12. Figure rendering may differ slightly with installed fonts.

Publication Readiness Checklist

Status: scientifically revised and locally reproducible; not yet submission-ready. This is not a guarantee of IEEE Access suitability or acceptance.

Completed

- Original mathematics data and original predictions reproduced.
- Title and claims restricted to retrospective classification.
- Same original holdout, grade exclusion, fixed threshold, and original default preserved.
- Baselines, ensembles, nested validation, ablations, uncertainty, calibration diagnostics, subgroups and stability executed.
- Official IEEE Access Word template obtained; editable manuscript and matching exported PDF prepared.
- Verified bibliography and factual AI-assistance acknowledgment added.
- Full numerical tables, figure sources, code, configurations and provenance retained.

Required author actions before submission

- Confirm spelling, exact institutional affiliation, authorship order, author contributions and corresponding-author email; no identity changes have been made.
- Provide approved biographies for all three authors; IEEE Access requires biographies below the references. They were not invented.
- Provide funding and conflict-of-interest declarations; neither absence of funding nor absence of conflicts is asserted.
- Obtain the institution's actual ethics determination for secondary analysis; public data do not establish an exemption.
- Review every AI-assisted section and script, approve the AI disclosure and verify all results.
- Publish a versioned, licensed code/data archive with a stable URL or DOI and replace the local-package availability statement with the actual archive citation. The authors must choose a code license and confirm redistribution rights.
- Confirm corresponding-author account, public populated ORCID, coauthor consent, and absence of simultaneous submission.
- Resolve the modest novelty and venue-fit concern: the work is an audit on one small historical benchmark, with no algorithmic novelty.
- Decide whether a fresh independent cohort and a timestamped prediction design are required to make the intended contribution sufficient. Without them, maintain all retrospective and exploratory qualifications.

- Run the journal's final file/metadata checks and select 3-10 keywords; confirm PDF/source content after inserting biographies and declarations.

Strongly recommended research before operational claims

Independent secondary-school cohort; prospective cutoff snapshots; time-separated evaluation; larger subgroup samples; support-capacity and cost elicitation; educator usability and intervention outcome studies. None can be inferred from current software checks.

Current IEEE Access requirements consulted

https://ieeeaccess.ieee.org/authors/submission-guidelines/ (accessed 1 November 2026). The page requires the official double-column single-spaced template, matching source/PDF files, AI disclosure and citation, author biographies, ORCID, and accurate references. No submission or communication with the journal has been performed.